

OSBORNE V. LOPES

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Summary

AI and Data Engineer with 3+ years of experience building production RAG pipelines, agentic multi-step workflows and semantic search systems using LangChain, LangGraph, Qdrant and Azure AI Search, alongside a background in data engineering across Snowflake, Kafka, Azure ADF and DBT. Skilled at taking AI systems from prototype to production, improving retrieval quality, reducing latency and integrating observability, guardrails and evaluation to ensure reliable outputs.

Experience

ServiceNow

Feb 2026 - Present

AI Engineer

MA

- Deployed a fine-tuned LLM chatbot on Azure Cloud Services handling 8,500+ daily employee queries, achieving 94% intent accuracy and reducing IT support ticket volume by 37% within 3 months.
- Led AI product go-to-market (GTM) initiatives by partnering with Product, Engineering, IT, and business stakeholders to define adoption strategies, pilot use cases, user feedback loops, and production rollout plans, accelerating enterprise adoption of AI-powered knowledge solutions.
- Built a FastAPI-based microservice with real-time vector search integration using Azure AI Search, enabling sub-200ms Similarity searches across 50M+ embedded vectors for document recommendation.
- Designed agent orchestration logic to manage multi-step state and conversation memory across workflows, evaluating LLM cost-performance trade-offs across model sizes and implementing self-critique, guardrails, and grounding mechanisms that improved response reliability by 30%.

Atlas SP

Jan 2025 - Aug 2025

AI Engineer Co-op

New York, NY

- Built and deployed a production RAG-based chatbot using Azure AI Search vector indexes, LangChain and FastAPI, enabling natural language querying of live financial data for internal teams and reducing repetitive analyst requests by 30%
- Improved retrieval quality by implementing hybrid search with a reranking layer to reprioritize retrieved passages before generation, integrating LangSmith for observability and monitoring key metrics like token usage and latency
- Built ADF batch-based and Kafka streaming-based ingestion pipelines from Azure Blob Storage to Snowflake, transforming data from raw to presentation layers and applying business logic through Alteryx workflows for downstream Tableau analytics
- Transformed raw data into production-ready datasets in Snowflake using complex views, stored procedures and Snowpipe for continuous data loading, supporting 10+ downstream Tableau dashboards
- Collaborated with team members through code reviews and PR merges to maintain pipeline quality, while partnering with business stakeholders to define KPIs and requirements for downstream Tableau dashboarding

Zyphron Industries

Jul 2021 - Aug 2023

Analytics Engineer

India

- Designed star-schema semantic models in Power BI with advanced DAX measures to track inventory, fulfillment, and supplier KPIs, earning the trust of logistics leadership across 5+ business teams.
- Analyzed customer delivery KPIs across 10+ enterprise clients using complex SQL window functions and incremental DBT models, applying A/B testing to reporting formats that reduced repeat customer complaints by 15%.
- Reduced inventory shortfalls by 20% by analyzing historical inventory and order data with SQL and Excel to identify peak-season demand patterns and anticipate potential stockouts.
- Automated ETL pipelines using Python (Pandas, NumPy, Scikit-Learn) to preprocess and validate 100k+ daily shipment, supplier, and fulfillment records, cutting manual prep time by 50% and enabling faster, cleaner data for downstream reporting.

Skills

Languages: Python, R, SQL, Pro*C

Databases & Warehousing: Snowflake, MongoDB, DynamoDB, Redshift

Cloud Platforms: Azure (ADF, Blob Storage, Synapse), AWS (S3, Glue, EMR, Lambda, Athena, Kinesis, IAM), GCP (BigQuery, DataFlow, GCS), Docker

Tools: Airflow, DBT, Apache Spark, Kafka, Databricks, Alteryx, Control-M, Git

Analytical Toolkit: Tableau, PowerBI, Looker, Google Data Studio, QuickSight, Excel

AI & Automation: RAG, Agentic AI, LangChain, LangGraph, HuggingFace, FastAPI, ChromaDB, Qdrant, PineCone, LLM, n8n, Prompt Engineering, Claude API, Gemini API, Model Context Protocol (MCP), Tool Calling, Vector Search, Embeddings, Reranking

Education

Northeastern University

Dec 2025

Master of Science, Information Systems

Boston, MA

- Coursework: Generative AI and Large Language Models, Big Data and Intelligent Analytics

University of Mumbai

Jun 2021

Bachelor of Technology, Information Technology

Mumbai, India

- Coursework: Data Mining and Business Intelligence, Database Management, Cloud Computing Services

Projects

RAG Job Assistant

May 2025

- Built a production-grade RAG pipeline indexing 5,000+ job postings into ChromaDB using HuggingFace sentence-transformer embeddings (all-MiniLM6-v2), enabling semantic search over job descriptions
- Architected a multi-agent retrieval system using LangGraph with ReAct-pattern agents for query decomposition, semantic search and answer synthesis, integrating LangChain and Groq API (Llama 3.3 70B) with prompt engineering to manage context windows, multi-turn interactions and output quality evaluation

NewsScope

Apr 2024

- Built an industry-specific news aggregator that scrapes, processes and surfaces real-time business news using Python (BeautifulSoup, Scrapy) and DBT
- Designed an end-to-end ETL pipeline on GCP using Dataflow for processing, Pinecone for vector embeddings and BigQuery as the data warehouse, enforcing data quality via Pydantic 2 and Pytest validations